

Tethered 6U CubeSat

What has flown before, how long the tether can be, and 3D photos

CUBICS 2026 · Stream 1 · Team meeting, October 2026

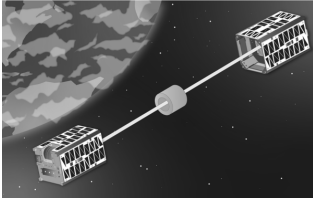
One tether, two 3U halves of 4 kg, orbit at 550 km.

What has flown before

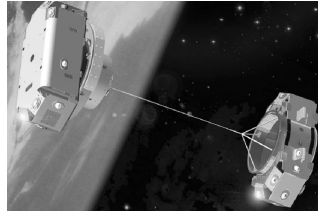
Mission	Year	Length	What happened
Gemini XI	1966	30 m	First tether between two spacecraft
SEDS-1	1993	20 km	Fully unrolled; the end mass fell back to Earth
SEDS-2	1994	19.7 km	The tether brake worked as designed
TSS-1R	1996	19.7 km	Tether broke (electrical spark)
TiPS	1996	4 km	Swing reduced from 40° to 7.5°
ATE _x	1999	22 m of 6.2 km	Tether went loose; unrolling stopped
YES2	2007	31.7 km	Longest object ever unrolled in orbit
TEPCE	2019	1 km	Two CubeSat halves pushed apart by a spring at 4 m/s

Tethers have flown for 60 years. When they fail, it is almost always while unrolling.

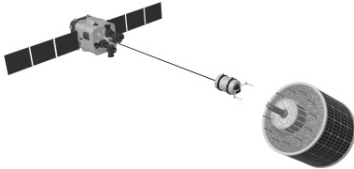
Previous tethered systems



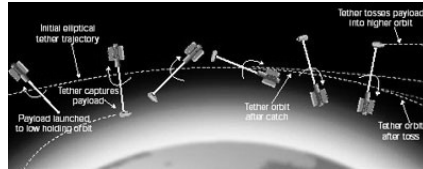
Two CubeSats joined by a tether



Two end bodies joined by a tether



Tether between a spacecraft and a target



Momentum exchange: catch and toss

What is already proven, and what is new

Already proven

- Splitting a CubeSat in two with a spring (TEPCE, 2019)
- Unrolling kilometres of tether with a brake (SEDS-2, TiPS)
- Keeping the pair pointed up and down by gravity alone
- Calming the swing with the winch

New in this proposal

- One tether with a winch that unrolls and reels in
- The tether angle measured at its anchor
- Tether reeled in between campaigns to last longer
- Earth photos from both halves

Why the tether stays tight

Orbit speed at radius $a = R_{\oplus} + h$:

$$n = \sqrt{\frac{\mu}{a^3}} = 1.0948 \times 10^{-3} \text{ rad/s}, \quad P = \frac{2\pi}{n} = 95.6 \text{ min.}$$

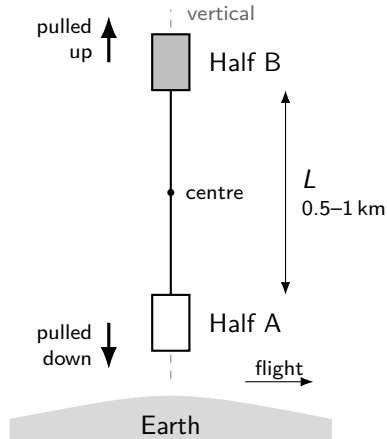
Gravity pulls the lower half a bit more, and the upper half a bit less, than its orbit needs. At a distance l from the centre the difference is

$$n^2(a + l) - \frac{\mu}{(a + l)^2} \simeq 3 n^2 l.$$

The tether makes up the difference, so it pulls with

$$T = 3 n^2 m_{\text{eq}} L, \quad m_{\text{eq}} = \frac{m_1 m_2}{m_1 + m_2} = 2 \text{ kg.}$$

$$T = 3.6 \text{ mN (500 m)} \quad T = 7.2 \text{ mN (1 km)}$$



It only works up and down

Motion of one half seen from the other (x up, y forward, z sideways):

$$\ddot{x} - 2n\dot{y} - 3n^2x = f_x, \quad \ddot{y} + 2n\dot{x} = f_y, \quad \ddot{z} + n^2z = f_z.$$

Direction	What happens	Tether
Up and down (x)	$+3n^2x$: the halves pull apart	tight
Forward and back (y)	no force at all	loose
Sideways (z)	$-n^2z$: the halves drift together	loose

A tilted tether swings back to vertical:

$$\ddot{\varphi} + 3n^2 \sin \varphi \cos \varphi = 0 \Rightarrow \text{swing period } \frac{2\pi}{\sqrt{3}n} = 55 \text{ min.}$$

A diagonal tether is a swing, not a position. Holding 30° at 1 km needs $3n^2 m_{\text{eq}} L^2 \sin \varphi \cos \varphi = 3.1 \text{ N m}$, about 3000 times what a CubeSat wheel gives.

500 m or 1 km with one tether: the numbers

	Formula	$L = 500 \text{ m}$	$L = 1 \text{ km}$
Pull on the tether	$3n^2 m_{\text{eq}} L$	3.6 mN	7.2 mN
Stretch	$TL/(EA)$	0.6 mm	2.3 mm
Tether mass, 0.2 mm thick	$\rho_c AL$	22 g	44 g
Area facing the air flow	$d L$	0.10 m ²	0.20 m ²
Time to unroll, slowly	$\ln(L/L_0)/\lambda$	3.2 h	3.5 h
Time to reel in, slowly	$\ln(L/L_0)/\lambda$	7.9 h	8.8 h
Time in orbit, tether out	drag model	$\approx 2.5 \text{ years}$	$\approx 1.6 \text{ years}$

$EA = 3142 \text{ N}$, $L_0 = 1 \text{ m}$, $\lambda = n/2$ out and $n/5$ in, average Sun. Tether reeled in: $\approx 6 \text{ years}$.

Once out, the length stays fixed

The winch sets the length. More pull does not make the tether longer; it only makes it tighter.

Stretch

$$\Delta L = \frac{T L}{EA}$$
$$\leq \mathbf{2.3 \text{ mm}}$$

at 1 km

Natural swing

$$2\pi/(\sqrt{3} n)$$

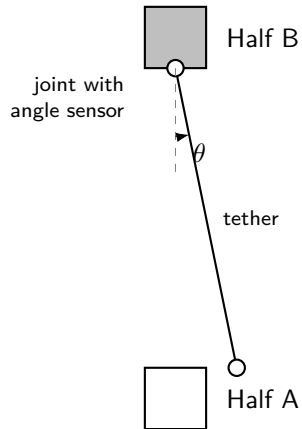
55 min

one slow swing around vertical

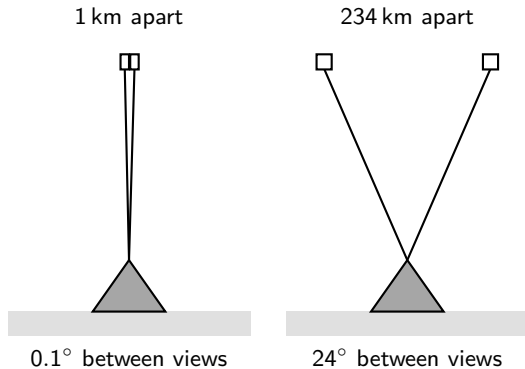
The pull depends only on the length: $T = 3n^2 m_{\text{eq}} L$. Unrolling faster only adds swing and more braking.

Angle between the halves: use the mechanism

- The tether already points from one half to the other.
- A free joint at each anchor, with angle sensors, reads that direction.
- It is a mechanical measurement: it does not drift and needs no extra electronics.
- When the two halves are joined, the angle is zero: that is our calibration.



3D photos need two very different viewing angles



A point of height h shifts between the photos by

$$p = h \frac{B}{H},$$

so the height error is

$$\sigma_h = \frac{\sigma_p}{B/H}.$$

B : distance between the two photos, $H = 550$ km, σ_p : how well the photos can be matched (1.5 m with 5 m pixels).

With the tether, the height error is about 1 km

Even if the tether were horizontal, the photos are only L apart: $B/H = L/H$ (5 m pixels).

$L = 1 \text{ km}$

$\sigma_h \approx 825 \text{ m}$

height error

$L = 500 \text{ m}$

$\sigma_h \approx 1.65 \text{ km}$

height error

1 km-high mountain

0.36 pixel

shift between the two photos

With a vertical tether it is worse: both cameras look along the same line, so there is no 3D at all.

What works: pointing the cameras

Option	B/H	Height error (5 m / 20 m pixels)	Where
A 12° ahead, B 12° back	0.43	3.5 m / 14 m	Under the track
Turning camera, $\pm 30^\circ$	0.40	3.8 m / 15 m	Any site, every 2–4 days
Tether, 1 km apart	1.8×10^{-3}	825 m / 3.3 km	Not usable

The 3D comes from the satellite moving 234 km along its orbit in 31 s, not from the tether.

Risks

Reeling in	Reeling in makes the swing grow. It must be slow and controlled: 7.9 to 8.8 h.
Time in orbit	Reel the tether in when not in use: about 6 years instead of 1.6 at 550 km. Let it out at the end to fall back faster.
One tether	One cut by a tiny piece of debris ends the mission. Use a braided line.
Space rules	A 1 km object is a bigger target for collisions. Check with the CSA that it is accepted.
Our estimates	Orbit times come from a simple model; a proper study (STELA or DRAMA) is needed.

In short

Previous work	Tethers have flown since 1966. Most failures happen while unrolling.
Tether length	Only up and down. 500 m to 1 km works: 3.6 to 7.2 mN of pull, about 3.5 h to unroll.
Angle between halves	Measured by the mechanism at the anchor, not by electronics that drift.
3D photos	Not with the tether: errors near 1 km. Yes by pointing the cameras: about 3.5 m.
Time in orbit	Reel the tether in when not in use: about 6 years instead of 1.6.

Decisions for the team

- 1 **Tether length:** 500 m or 1 km, reeled in between campaigns.
- 2 **Angle between halves:** joint with angle sensors at each anchor.
- 3 **3D photos:** drop 3D from the tether; point the cameras instead.
- 4 **Questions to the CSA by 9 November:** launch height, is the tether accepted, how it falls back.
- 5 **Study:** proper orbit-time study with STELA or DRAMA.